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### Mechanism for electronic adjustment of flows in fixed displacement pump

#### Abstract

An electronic angle adjustment mechanism for a pump and a motor generally includes a base and an actuating motor. The base has an upper portion mounted to a motor, a lower portion mounted to a pump, and a hinge for pivotably moving the upper base portion in relation to the lower base portion. The actuating motor is mounted on an attachment plate that is attached to the upper base portion. The actuating motor is connected to a worm screw that engages an arcuate member attached to the lower portion to pivot about the hinge and change the angle between the upper and lower base portions.

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## Background/Summary

(1) This divisional application claims priority from U.S. patent application Ser. No. 17/603,617, filed Oct. 14, 2021, which claims priority from International Application No. PCT/US2020/44252, filed Jul. 30, 2020, which claims priority from U.S. Provisional Patent Application Ser. No. 62/881,086, filed Jul. 31, 2019, the disclosures of which are incorporated herein in their entirety for all purposes.

### FIELD OF INVENTION

(1) The present invention relates to pumps used to dispense small amounts of fluids at accurate flow rates. In particular, the invention relates to a mechanism that electronically adjusts the dispensing of fluids from a pump at low flow rates.

### BACKGROUND

(2) A family of valve-less pumps, which have at their heart special mounting means, commonly referred to as a base, interposed between a drive motor and a pump head, is known in the art. These bases are typically injection molded plastic and incorporate a living hinge separating an upper base portion from a lower base portion. The upper base portion can be tilted with respect to the lower base portion by flexure of the living hinge. The relative angle between the upper and lower base portions establishes the pump output volume per revolution. This entire mechanism was previously described in commonly owned U.S. Pat. Nos. 5,020,980 and 4,941,809, and U.S. Patent Application Publication No. 2016/0245275, each of which is incorporated herein in its entirety.

(3) Conventionally, the method for adjusting and setting the angle is accomplished by means of an adjusting screw engaging with pivot pins in the two portions of the base, which are positioned on the opposite side of the central axis of the base. Certain applications require pumps with the same target output per revolution. This was accomplished by substituting fixed linkage means for the adjustable screw and pivot pins. The fixed links are injection molded from plastic resin and the tooling used to mold these links allows for different lengths to be produced such that different target pump displacements can be routinely produced. An eccentric bushing providing a combination of the benefits of an adjusting screw and a fixed link is disclosed in commonly owned U.S. Patent Application Publication No. 2016/0245275.

(4) These traditional methods for changing the output volume per revolution by adjusting the angle between the upper base portion and lower base portion have all required manual adjustment. This has generally made conventional pumps only convenient for use at a single output volume per revolution.

(5) However, there are applications where it would be beneficial to be able to electronically adjust the output volume per revolution. This would allow an electronic system to adjust these pumps without manual intervention. U.S. Pat. No. 7,708,535 discloses a method for electronic adjustment of the angle of the base. However, the device disclosed in this patent uses rigid members to translate linear motion to angular motion. This leads to varying angular movement relative to linear movement, which leads to a complex relationship when defining the linear motion required to adjust the angle between the two portions of the base.

(6) Moreover, due to the nature of the mechanism linking the piston to the motor shaft, the output volume is not a constant flow rate when the motor is rotated at a constant speed. Instead, the flow rate through the pump head is sinusoidal with the dispense to the outlet port being the positive portion of the sine wave and the aspirate from the inlet port being the negative portion of the sine wave.

(7) However, there are applications where the sinusoidal nature of the dispense is not acceptable and a constant flow rate is desired. In these cases, a traditional syringe pump is generally favored for the constant flow rate it can easily provide.

(8) There are also applications where a pump is used to dispense a small volume. This sometimes means that it takes a significant length of time to prime the line from the fluid source to the pump and to the dispense tip.

(9) In certain cases, a pump is used to aspirate a fluid into a probe tip and dispense portions of the aspirated fluid into other receptacles. A fixed displacement pump can be used for these cases by rotating the motor in the reverse direction to aspirate. However, due to the design of the fixed displacement pump, the aspirate volume may not be the same as the calibrated dispense volume.

(10) Another drawback with traditional syringe pumps is that a linear actuator is used to move the plunger to pull fluid into and push fluid out of a barrel. The accuracy of a syringe pump is generally tied to the size of the syringe barrel. The larger a syringe barrel, the lower its accuracy and precision. In order to have high accuracy at smaller volume dispenses or aspirates, a smaller barrel must be used. This is due to the smallest reliable increment of linear distance travelled in a syringe pump being related to a volume of fluid being moved. As the barrel size grows, this increment of linear distance relates to a larger volume of fluid being moved.

(11) Still another drawback with pumps of the prior art relates to the need for priming such pumps. In order to decrease priming time and limit use of the syringe pump as much as possible, some systems include a priming pump with a syringe pump. The priming pump fills the lines quicker than the syringe pump and also limits the use of the syringe pump in order to increase the time between required maintenance of the syringe pump.

(12) Accordingly, it would be desirable to provide a means for remote adjustment of output volume per revolution of a fixed displacement pump. It would be further desirable to provide a mechanism capable of overcoming the restrictions of sinusoidal output of a fixed displacement pump and also capable of varying output volume per revolution. It would also be desirable to overcome issues of varying aspirate volumes relative to dispense volumes in a fixed displacement pump and to overcome accuracy restrictions related to syringe pump barrel sizes, while also incorporating priming capabilities.

## SUMMARY

(13) In a first embodiment of the present invention, an electronic angle adjustment mechanism for a pump and a motor is provided. The mechanism generally includes a base, a linear actuator and a flexible member. The base has a motor flange for mounting a motor, a pump flange opposite the motor flange for mounting a pump and a hinge or hinge assembly disposed between the motor flange and the pump flange. The pump flange can be integrally formed as part of a collar that is attached to the pump housing or can be formed as part of the pump housing. The linear actuator is mounted to one of the motor flange or the pump flange of the base and the flexible member has a proximal end attached to the linear actuator and a distal end opposite the proximal end connected to the other of the motor flange or the pump flange of the base. When actuated, the linear actuator drives the flexible member in a curved path causing the motor flange and the pump flange to pivot with respect to each other about the hinge, thereby changing an angle between the motor flange and the pump flange.

(14) The electronic angle adjustment mechanism can also include a cam block mounted to one of the motor flange or the pump flange, wherein the cam block has a curved support surface for guiding the flexible member in the curved path. An attachment plate can be mounted between the motor flange and the motor. The attachment plate extends outwardly from the motor parallel to the face of the motor flange and is sized to accommodate the mounting of the electronic adjustment mechanism. Preferably, the attachment plate is integrally formed as part of the motor flange. The curved support surface has a radius of curvature about a pivot point of the base hinge defined by the distance from the pivot point to the connection point of the flexible member with the other of the motor flange or the pump flange.

(15) In the first embodiment, the angle adjustment mechanism preferably includes a roller bearing adjacent the cam block. The roller bearing presses the flexible member against the curved surface

of the cam block.

(16) The flexible member may comprise a spring steel material such that the flexible member is bendable for transitioning a linear motion of the linear actuator to a pivoting motion of the motor flange and the pump flange with respect to one another.

(17) In another aspect of the first embodiment of the present invention, a motor and pump assembly is provided. The motor and pump assembly generally includes a base, a motor, a pump, a linear actuator and a flexible member. The base includes a motor flange, a pump flange opposite the motor flange and a hinge disposed between the motor flange and the pump flange. The motor is mounted to the motor flange of the base, and has a shaft rotatable about a rotation axis. The pump is mounted to the pump flange of the base, and has a piston rotatable about a rotation axis and linearly translatable along the rotation axis, wherein the pump piston is coupled to the motor shaft. The linear actuator is mounted to one of the motor flange or the pump flange of the base, and the flexible member has a proximal end attached to the linear actuator and a distal end opposite the proximal end connected to the other of the motor flange or the pump flange of the base. When actuated, the linear actuator drives the flexible member in a curved path causing the motor flange and the pump flange to pivot with respect to each other about the hinge thereby changing an angle between the rotation axis of the motor shaft and the rotation axis of the pump piston about the hinge.

(18) In one aspect of the present invention, the linear actuator includes a drive rod movable along a linear axis, and a drive rod coupler attached to a distal end of the drive rod, wherein the flexible member is attached to the drive rod coupler. In this aspect, the linear actuator is preferably mounted to the motor flange and the drive rod extends parallel with the rotation axis of the motor shaft. The linear actuator can be a DC, AC, or a brushless DC motor, more preferably a stepper motor.

(19) In another aspect of the present invention, a method for adjusting the angular orientation between a motor shaft of a motor and a pump piston of a pump is provided. The method generally includes providing a base between the motor and the pump, wherein the base includes a motor flange for mounting the motor, a pump flange opposite the motor flange for mounting the pump and a hinge assembly disposed between the motor flange and the pump flange, and driving a flexible member in a curved path against one of the motor flange or the pump flange with a linear actuator mounted to the other of the motor flange or the pump flange, thereby changing an angle between the motor shaft and the pump piston about the hinge assembly.

(20) In a second embodiment of the electronic adjustment mechanism, the pump and motor are the same as in the first embodiment described above. The base is formed by an upper base portion and a lower base portion that are pivotably connected by a hinge or hinge assembly but a different electronic adjustment mechanism is used. In the second embodiment, the attachment plate extends outwardly from the motor and a sidewall extends downwardly. An electric motor, preferably a DC, AC, or brushless DC motor, more preferably a stepper motor, is attached to the outside of the sidewall and the motor shaft passes through the sidewall. A gear wheel with a plurality of teeth is attached to the distal end of the motor shaft. A collar is attached to the lower base portion. The collar fits around the outside of the lower base portion and is attached by a clamp, screws, bolts, an adhesive, or other known fastening devices. The collar can also be integrally formed as part of the lower base portion or pump housing and it can also have a flange extending outwardly from at least part of the exterior surface. On one side of the collar, the lower base portion is attached to the upper base portion by the hinge. Opposite the hinge, a bracket having two parallel members with a slot in between extends outwardly from the collar. On the distal ends of the two parallel members an arcuate member is attached between the two parallel members. The arcuate member curves inwardly towards the collar and has a concave surface with a plurality of teeth. The plurality of teeth on the gear wheel engage the plurality of teeth on the arcuate member and the motor controls the pivotal movement of the upper base portion in relation to the lower base portion.

(21) In a third embodiment of the electronic adjustment mechanism, the pump and motor are the

same as the first embodiment described above. The base is formed by an upper base portion and a lower base portion that are pivotably connected by a hinge or hinge assembly but a different electronic adjustment mechanism is used. In the third embodiment, the attachment plate extends outwardly from the motor and a sidewall extends downwardly. An electric motor, preferably a DC, AC, or brushless DC motor, more preferably a stepper motor, is attached to the outside of the member and the motor shaft passes through the sidewall. A gear wheel with a plurality of teeth is attached to the distal end of the motor shaft. A collar, as described above, is attached to the lower base portion. One side of the collar is attached to the upper base portion by the hinge. Opposite the hinge, a bracket having two parallel members with a slot in between extends outwardly from the collar. On the distal ends of the two parallel members an arcuate member is attached between the two parallel members. The arcuate member curves outwardly away from the collar and has a convex surface with a plurality of teeth. The plurality of teeth on the gear wheel engage the plurality of teeth on the arcuate member and the motor controls the pivotal movement of the upper base portion in relation to the lower base portion.

(22) In a fourth embodiment of the electronic adjustment mechanism, the pump and motor are the same as the first embodiment described above. The base is formed by an upper base portion and a lower base portion that are pivotably connected by a hinge or hinge assembly but a different electronic adjustment mechanism is used. In the fourth embodiment, the attachment plate extends outwardly from the motor. A motor, preferably a DC, AC, or brushless DC motor, more preferably a stepper motor, is mounted on the attachment plate with the motor shaft extending downwardly through the plate towards the pump. A worm screw is attached to the distal end of the motor shaft. A collar, as described above, is attached to the lower base portion. One side of the collar is attached to the upper base portion by a hinge assembly. Opposite the hinge assembly, a bracket having two parallel members with a slot in between extends outwardly from the collar. On the distal ends of the two parallel members, an arcuate member is attached between the two parallel members. The arcuate member curves outwardly away from the collar and has a convex surface with a plurality of teeth. The plurality of teeth on the worm screw engage the plurality of teeth on the arcuate member and the motor controls the pivotal movement of the upper base portion in relation to the lower base portion.

(23) Thus, the invention utilizes a linear actuator to allow electronic adjustment of the angle between the pump piston and the motor shaft. The linear actuator is mounted to the upper base portion and adjustably connected to the lower base portion. With this invention, the angle is adjustable electronically instead of manually.

(24) By facing the piston flat to a port and varying the angle by means of the linear actuator, the pump can “syringe” fluid and dispense or aspirate at a near constant flow rate. When the linear actuator is extended, this will increase the angle between the portions of the base and the pump will aspirate through the active port. When the linear actuator is retracted, this will decrease the angle between the portions of the base and the pump will dispense through the active port.

(25) With the ability to electronically adjust the angle, the angle can be manually or automatically adjusted to operate at one of several output volumes per revolution. For example, a large angle would be used for a high output volume per revolution for priming or flushing the fluid circuit. Then, the angle would be electronically adjusted to a small angle for a low output volume per revolution for small volume critical dispenses. With the ability to “syringe” fluid, a predictable and accurate aspirate and dispense volume can be achieved.

(26) By varying the angle between the piston flat and the active port, varying barrel sizes can be achieved. This means that a single pump can be used to dispense fluids at rates equivalent to pumps with a large barrel size and pumps with a small barrel.

(27) In still another aspect, this invention could be used like a traditional pump to prime the fluid circuit, and then operated like a syringe pump. This eliminates the need for two separate pumps and combines the syringe pump with the priming pump.

(28) Features of the disclosure will become apparent from the following detailed description considered in conjunction with the accompanying drawings. It is to be understood, however, that the drawings are designed as an illustration only and not as a definition of the limits of this disclosure.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a conventional motor/pump connection utilizing adjustable flow angle hardware, according to the prior art.
- (2) FIG. 2 is a perspective view of a conventional motor/pump connection utilizing a fixed link, according to the prior art.
- (3) FIG. 3 is a cross-sectional view of a liquid pump according to the prior art.
- (4) FIG. 4 is a perspective view of a motor/pump connection utilizing an electronic, angle adjustment mechanism according to a first embodiment of the present invention.
- (5) FIG. 5 is a front view of the motor/pump connection utilizing an electronic adjustment mechanism shown in FIG. 4.
- (6) FIG. 6 is a cross-sectional view of the motor/pump connection utilizing an electronic adjustment mechanism taken along the line 6-6 in FIG. 5.
- (7) FIG. 7 is a perspective view of a motor/pump connection utilizing an internal gear, electronic angle adjustment mechanism according to a second embodiment of the present invention.
- (8) FIG. 8 is a perspective view of a motor/pump connection utilizing an external gear, electronic angle adjustment mechanism according to a third embodiment of the present invention.
- (9) FIG. 9 is a perspective view of a motor/pump connection utilizing a worm screw, electronic angle adjustment mechanism according to a fourth embodiment of the present invention.

### DETAILED DESCRIPTION

- (10) FIG. 1 shows a conventional prior art motor **10** connected to a pump **12** via a base **14**. The motor **10** has a shaft that rotates about a rotational axis and the pump has a piston that also rotates about a rotational axis and also translates in the direction of the rotational axis. The shaft of the motor is coupled to the piston of the pump so that rotation of the motor shaft will cause rotation of the pump piston. Also, by tilting the rotational axis of the pump piston with respect to the rotational axis of the motor shaft, rotation of the motor shaft will also cause linear translation of the pump piston in a manner that is described in further detail below. A pump and motor support arrangement of this type is shown and described in commonly owned U.S. Pat. Nos. 4,941,809 and 5,020,980, the specifications of which are incorporated herein by reference in their entirety for all purposes.
- (11) FIG. 1 shows one prior art embodiment of an adjustable base **14**, which includes a flange attached to the motor **10** and an opposing or mating flange attached to the pump **12**. Between the two flanges is a flexible living hinge, which allows angular pivoting of the flanges with respect to the hinge. Opposite the hinge are two bosses, between which adjustable flow angle hardware is provided. In the embodiment shown in FIG. 1, the adjustable flow angle hardware is in the form of a screw and nut arrangement connected between pivot pins inserted in the respective bosses of the base. Rotation of the nut with respect to the screw selectively lengthens or shortens the length between the pivot pins of the bosses, thereby adjusting the angle of the motor flange with respect to the pump flange.
- (12) FIG. 2 shows an alternative embodiment of a prior art motor/pump connection of the prior art utilizing a base, similar to the base shown in FIG. 1, but utilizing a fixed link provided between the opposing bosses. Specifically, the base **14** shown in FIG. 2, again includes a motor mounting flange and a pump mounting flange on opposite sides of a flexible living hinge. Opposite the hinge are opposed bosses between which a fixed link is provided to set the angle between the pump and the

motor. The length of the fixed link is selected based on the desired volumetric flow produced by the pump. In certain applications, a variety of fixed links of differing lengths can be provided to adjust the volume of the pump in a predetermined range.

(13) Referring now to FIG. 3, this prior art pump and motor arrangement operates as follows. The pump **12** generally includes a pump housing **101** and a piston **118**. The pump housing **101** includes a plastic pump casing **102** having an inlet port **104** and an outlet port **106**. The pump casing **102** defines a cylindrical chamber **108** having an open end **110**. Received in the cylindrical chamber **108** is a ceramic piston liner **112** having a central longitudinal bore **114** and a transverse bore **116** communicating with the longitudinal bore **114**. The transverse bore **116** includes a liner inlet port **116a** fluidly communicating with the inlet port **104** of the pump casing **102** and a liner outlet port **116b** fluidly communicating with the outlet port **106** of the pump casing so that a liquid can be pumped from the inlet port **116a**, through the liner, to the outlet port **116b** in a manner described below.

(14) The pump piston **118** is axially and rotatably slidable within the central bore **114** of the piston liner **112**. One end of the piston **118** extends out of the open end **110** of the pump casing **102** and includes a coupling **120** for engagement with the shaft of the motor **10**. At its opposite end, the piston **118** is formed with a relieved or “cutout” portion **122** disposed adjacent the transverse bore **116** of the pump liner. As described below, the relieved portion **122** is designed to direct fluid into and out of the pump **12**.

(15) A seal assembly **124** is provided at the open end **110** of the pump casing **102** to seal the piston **118** and the pump chamber **108**. The seal assembly **124** is retained at the open end **110** of the pump casing **102** by a gland nut **126** having a central opening **128** to receive the piston **118**. The gland nut **126** is attached to the pump casing **102** with a threaded connection **130**.

(16) In operation, the motor **10** drives the piston **118** to axially translate and rotate within the central bore **114** of the piston liner **112**. In order to draw liquid into the transverse bore **116** from the inlet port **104**, the piston **118** is rotated as required to align the relieved portion **122** with the liner inlet port **116a**. The piston **118** is then drawn back as required to take in the desired volume of liquid into the central bore **114** of the pump liner **112**. Withdrawal of the piston **118** produces a negative pressure within the liner inlet port **116a** of the transverse bore **116**, which draws in liquid from the casing inlet port **104**. The piston **118** is then rotated to align the relieved portion **122** with the liner outlet port **116b**. Finally, the piston **118** is driven forward the required distance to force liquid into the outlet port **116b** of the transverse bore **116** to produce the desired discharge flow.

(17) Thus, each rotation of the motor shaft rotates the piston of the pump. Due to the angular orientation between the pump and the motor, each rotation of the motor shaft further causes the pump piston to reciprocate in the axial direction to alternately draw in and push out fluid to transfer fluid between an inlet and an outlet of the pump. The amplitude of the piston stroke determines the volume of the fluid delivered between the inlet and the outlet of the pump. By varying the angle of the pump with respect to the motor, the stroke of the piston is adjusted, thereby adjusting the volume of the fluid transferred between the inlet and the outlet.

(18) In such prior art pump and motor arrangements, the angle of the pump **12** with respect to the motor **10** is adjustable via the base **14** to provide a desired volumetric flow of the pump with each rotation of the motor shaft. Therefore, it is desirable to provide a base **14** which is adapted for adjusting the angle between the axis of the pump and the motor shaft.

(19) As used herein, a “stepper motor,” also known as step motor or stepping motor, is an electric motor that divides a full shaft rotation into a number of steps of essentially uniform magnitude when driven from a sequentially switched DC power supply.

(20) As used herein, the term “worm drive” is a gear arrangement in which a worm or worm screw meshes with an arcuate (i.e., curved) member with a plurality of teeth. The worm screw and arcuate member are arranged in parallel along their longitudinal axes and the threads of the worm screw engage the teeth of the arcuate member. Rotation of the worm screw in a clockwise direction

causes the arcuate member to move in a first direction and rotation of the worm screw in a counterclockwise direction causes the arcuate member to move in the opposite direction.

(21) As used herein, the terms “hinge,” “hinge assembly,” and “living hinge” refer to a movable joint or mechanism having one or more components, which connect(s) the upper base portion and lower base portion to change the angular relationship between their longitudinal axes.

(22) As used herein, the term “living hinge” refers to a type of hinge made from an extension of the parent material (typically plastic). The living hinge “bridge” is the thin section of plastic that acts as a connection between two larger plastic sections, i.e., the upper base portion and the lower base portion. Preferably, the upper and lower base portions and the living hinge “bridge” will be made of one continuous piece of plastic. Since it is very thin and typically made from a flexible plastic, the living hinge is also able to rotate about one axis 180 degrees or more.

(23) Referring now to FIGS. 4-6, an adjustable pump and motor assembly 20 with an angle adjustment actuator 60 according to a first embodiment of the present invention is shown. The adjustable pump and motor assembly 20 includes a conventional motor 22 connected to a fixed displacement pump 24 (as described above with reference to FIG. 3) via a base 26 with a pivotably connected upper base portion 46 and a lower base portion 48. The motor 22 has a shaft 28 that is connected to a spindle coupling 32 and the shaft 28 rotates the spindle coupling 32 about a rotational axis. The pump 24 has a piston 30 that also rotates about a rotational axis and also translates in the direction of its rotational axis. One end of the piston 30 is connected to the spindle coupling 32.

(24) The shaft 28 of the motor 22 is coupled to the piston 30 of the pump 24 via the spindle coupling 32 so that rotation of the motor shaft 28 will cause rotation of the pump piston 30. Also, by tilting the rotational axis of the pump piston 30 with respect to the rotational axis of the motor shaft 28, rotation of the motor shaft 28 will also cause linear translation of the pump piston 30 and increase or decrease the volume of the chamber 35 at the distal end of the piston 30.

(25) The end of the pump piston 30 closer to the motor shaft 28 is attached to a pin 34 that is perpendicular to the pump piston 30 and connected to a spherical bearing 36. The spherical bearing 36 is retained or captured in a hollow portion of the spindle coupling 32. When the spindle coupling 32 is rotated by the motor shaft 28, the spherical bearing 36 and pin 34 assembly translates the rotational movement of the spindle coupling 32 to the pump piston 30. Rotation of the spindle coupling 32 rotates and reciprocates the pump piston 30 inside the cylinder 38 of the pump 24 in a linear direction along the axis of the pump piston 30. As the pump piston 30 moves linearly, the spherical bearing 36 rotates in the hollow of the spindle coupling 32. The reciprocal rotation of the pump piston 30 over a 180-degree arc switches the piston flat 44 between a first position facing the first port 40 and a second position facing the second port 42. In the first position, the piston flat 44 allows fluid to flow from the first port 40 into the chamber 35. As the pump piston 30 rotates 180-degrees, the first port 40 is closed off and the piston flat 44 moves to the second position and dispenses the fluid from the chamber 35 through the second port 42. As the pump piston 30 reciprocally rotates in the cylinder 38 between opposing ports 40, 42, the piston flat 44 is open to only one port 40, 42 at a time.

(26) The port 40, 42 that is open to the piston flat 44 is considered the active port. The reciprocating motion pulls fluid in from and pushes fluid out of the active port. The reciprocation and rotation is timed to pull fluid in from one port and push fluid out of the opposite port. Preferably, the piston flat 44 reciprocates by rotating about 180 degrees between the ports 40, 42. Modifying the angle that the pump piston 30 is held relative to the motor shaft 28 adjusts the volume in the chamber 35 at the bottom of the pump piston 30 so that the output volume per revolution can be calibrated to a desired output volume.

(27) As also discussed above, the angle between the axis of the pump piston 30 and the motor shaft 28 is determined by means of the base 26 having an upper base portion 46 and a lower base portion 48 pivotably connected to one another via a hinge 50. The upper base portion 46 has a flange 52

that attaches to the motor **22**, and the lower base portion **48** has a flange **54** that holds the pump head **24** that houses the piston **30** and cylinder **38**. The hinge **50** allows the upper base portion **46** to be tilted relative to the lower base portion **48** in a direction indicated by arrow **47** in FIG. **4**. Typically, the base **26**, including the upper base portion **46** and lower base portion **48**, are injection molded together with a living hinge **50**. However, it is within the scope of the invention for these portions to be molded separately with a pinned hinge instead.

(28) The piston **30** extends into the cylinder **38** and forms a chamber **35** between the distal end of the piston **30** and the bottom of the cylinder **38**. The volume of the chamber **35** changes as the piston **30** travels up and down in the cylinder **38**. Adjusting the angle between the axis of the pump piston **30** and the motor shaft **28** adjusts the travel distance of the piston **30** and determines the maximum volume of the chamber **35** and the flow rate.

(29) Adjustment of the angle between the motor shaft **28** and the pump piston **30** is achieved with an electronic adjustment mechanism **59** according to a first embodiment of the present invention shown in FIGS. **4-6**. The electronic adjustment mechanism **59** includes a linear actuator **60** attached to one of the flanges of the base **26**. FIGS. **4-6** are directed to a first embodiment of the present invention, wherein a linear actuator **60** attached to the motor flange **52** of the upper base portion **46**. However, it is conceivable for the actuator **60** to be attached to the opposite pump flange **54**, wherein the arrangement of the remaining associated components described herein would be reversed.

(30) The linear actuator **60** is preferably an electronic device capable of translating a linear actuator drive rod **62** in precise increments along a linear axis **64** extending parallel to the rotational axis of the motor shaft **28**. One type of linear actuator for use in the present invention is known in the art as a captive nut linear actuator.

(31) The motor flange **52** on the upper base portion **46** is preferably attached to the motor **22** by an attachment plate **66**. The attachment plate **66** extends outwardly from the motor **22** and is sized and shaped to allow mounting of the linear actuator **60** of the electronic angle adjustment mechanism **59** to an upper surface **68** of the attachment plate **66**. The mounting of the linear actuator **60** and the motor **22** on the upper surface **68** of the attachment plate **66** and mounting of the motor flange **52** on a lower surface **70** of the attachment plate **66** can be accomplished with conventional fasteners, such as bolts with threaded connections in respective components. Preferably, the attachment plate **66** extends outwardly from the motor **22** and is formed from a single sheet of metal and shaped to accommodate the electronic angle adjustment mechanism **59**.

(32) Attached to a distal end of the linear actuator drive rod **62** of the linear actuator **60** is a drive rod coupler **72**. The drive rod coupler **72** extends outwardly from the linear actuator **60** in the axial direction along the longitudinal axis **64**. The drive rod coupler **72** further extends axially through an opening provided in the attachment plate **66** between the upper and lower surfaces. Attached to a distal end of the drive rod coupler **72**, opposite the drive rod **62** is a flexible member **74**.

(33) The flexible member **74** is preferably made from a material having the strength to transfer the linear force imparted by the drive rod **62** along its longitudinal axis **64**, yet flexible enough to allow for some slight bending, as will be discussed further below. A suitable material for the flexible member, for example, is spring steel.

(34) The flexible member **74** has a first end attached to the distal end of the drive rod coupler and a second end, opposite the first end, connected to the lower flange **54** of the base **26**. Thus, linear motion of the linear actuator drive rod **62** will cause linear motion of the flexible member **74** in the same direction. Because the linear actuator **60** is connected to the upper base portion **46** and the flexible member **74** is connected to the lower base portion **48**, linear motion of the flexible member **74** will cause the lower base portion **48** to pivot with respect to the upper base portion **26** about the hinge **50**.

(35) The flexible member **74** initially extends from the drive rod coupler **72** in a direction along the linear axis **64** of the linear actuator drive rod **62**. However, the flexible member **74** is permitted to

begin to bend at a point along the longitudinal axis **64** beyond the drive rod coupler **72**. Such bending of the flexible member **74** is desirable to compensate for the arc shaped path of travel of the end of the lower flange **54** opposite the base hinge **50**.

(36) The bending of the flexible member **74** can be facilitated by a cam block assembly **76** and a roller bearing assembly **78**. The cam block assembly **76** includes a bracket **80** mounted to the lower flange **54** of the base **26** opposite the base hinge **50**. Any attachment means can be used. For example, a conventional screw fastener engaged in a threaded hole formed in the lower flange **54** will be sufficient.

(37) The cam block assembly **76** further includes a cam block **82** supported by the bracket **80**. The cam block **82** has a curved support surface **84** facing the flexible member **74**. The curved support surface **84** of the cam block **82** has a radius of curvature about the pivot point of the base hinge **50** defined by the distance from the pivot point to the intersection point of the flexible member **74** with the lower flange **54** of the base **26**. With the flexible member **74** bearing against the curved support surface **84** of the cam block **82**, the flexible member **74** will traverse a curved path coinciding with the path of the distal end of the lower flange **54** about the base hinge **50**.

(38) The roller bearing assembly **78** includes a bracket **86** mounted to the attachment plate **66**. The bracket **86** rotatably supports a roller bearing **88** positioned opposite the cam surface **84** of the cam block **82**. In this regard, the roller bearing **88** can be rotatably mounted on a pin fixed to the roller bearing assembly bracket **86**. The roller bearing **88** here is used to help constrain the flexible member **74** against the curved support surface **84**. One or more springs (not shown) could also be included with the roller bearing assembly **78** to provide an ongoing bias on the roller bearing **88** for pressing the flexible member **74** against the cam block **82**. Without the roller bearing **88**, the flexible member **74** would only be constrained by the drive rod **62** and would therefore, be susceptible to bending outwardly.

(39) As can be appreciated from the description above, at least some embodiments of the present invention include a controller that is coupled to the motor **22** and the linear actuator **60** via respective electrical lines **90**, **92**. One such example of a controller is a computer device that enables dynamic control of the linear actuator **60** and causes the electronic adjustment mechanism **59** to be precisely and repeatably modified. As such, the volume of fluid dispensed is extremely accurate, repeatable, and dynamic. One skilled in the art will appreciate that the invention may be practiced by one or more computing devices and in a variety of system configurations, including in a networked configuration.

(40) A second embodiment of the electronic adjustment mechanism **259** of the present invention is shown in FIG. **7**. The attachment plate **266** is mounted between the motor **222** and a motor flange **252** on an upper base portion **246** and extends outwardly on one side of the motor **222**. The upper base portion **246** and a lower base portion **248** connected to the pump **224** are pivotably connected by a hinge **250**. A sidewall **268** on one side of the attachment plate **266** extends downwardly from the motor **222** towards the pump **224**. An electric motor **260** is attached to the outside of the sidewall **268** and the motor shaft **262** passes through the sidewall **268**. A gear wheel **274** with a plurality of teeth **276** is attached to the distal end of the motor shaft **262**.

(41) A collar **254** is attached to the lower base portion **248** and one side of the collar **254** is attached to the upper base portion **246** by the hinge **250**. Opposite the hinge **250**, a bracket **278** having two parallel members **280**, **282** extends outwardly from the collar **254**. On the distal end of the two parallel members **280**, **282**, an arcuate member **284** is attached between the two parallel members **280**, **282**. The arcuate member **284** curves inwardly towards the collar **254** and has a plurality of teeth **286** on the concave, inward surface. The plurality of teeth **276** on the gear wheel **274** engage the plurality of teeth **286** on the arcuate member **284** and the motor **260** controls the pivotal movement of the upper base portion **246** in relation to the lower base portion **248**.

(42) A third embodiment of the electronic adjustment mechanism **359** of the present invention is shown in FIG. **8**. The attachment plate **366** is mounted between the motor **322** and a motor flange

**352** on an upper base portion **346** and extends outwardly on one side of the motor **322**. The upper base portion **346** and a lower base portion **348** connected to the pump **324** are pivotably connected by a hinge **350**. A sidewall **368** on one side of the attachment plate **366** extends downwardly from the motor **322** towards the pump **324**. An electric motor **360** is attached to the outside of the sidewall **368** and the motor shaft **362** passes through the sidewall **368**. A gear wheel **374** with a plurality of teeth **376** is attached to the distal end of the motor shaft **362**.

(43) A collar **354** is attached to the lower base portion **348** and one side of the collar **354** is attached to the upper base portion **346** by the hinge **350**. Opposite the hinge **350**, a bracket **378** having two parallel members **380**, **382** extends outwardly from the collar **354**. On the distal end of the two parallel members **380**, **382**, an arcuate member **384** is attached between the two parallel members **380**, **382**. The arcuate member **384** curves outwardly away from the collar **354** and has a plurality of teeth **386** on the convex, outward surface. The plurality of teeth **376** on the gear wheel **374** engage the plurality of teeth **386** on the arcuate member **384** and the motor **360** controls the pivotal movement of the upper base portion **346** in relation to the lower base portion **348**.

(44) A fourth embodiment of the electronic adjustment mechanism **459** of the present invention is shown in FIG. **9**. The attachment plate **466** is mounted between the motor **422** and a motor flange **452** on an upper base portion **446** and extends outwardly on one side of the motor **422**. The upper base portion **446** and a lower base portion **448** are pivotably connected by a hinge **450**. A motor **460** is mounted on the attachment plate **466** with the motor shaft **462** extending downwardly through the plate **466** towards the pump **424**. A worm screw **474** with a continuous spiral thread **476** is attached to the distal end of the motor shaft **462**.

(45) A collar **454** is attached to the lower base portion **448** and one side of the collar **454** is attached to the upper base portion **446** by a hinge **450**. Opposite the hinge **450**, a bracket **478** having two parallel members **480**, **482** extends outwardly from the collar **454**. On the distal end of the two parallel members **480**, **482**, an arcuate member **484** is attached between the two parallel members **480**, **482**. The arcuate member **484** curves outwardly away from the collar **454** and has a plurality of teeth **486** on the convex, outward surface. The continuous spiral thread **476** on the worm screw **474** engages the plurality of teeth **486** on the arcuate member **484** and the motor **460** controls the pivotal movement of the upper base portion **446** in relation to the lower base portion **448**.

(46) Embodiments of the present invention embrace one or more computer readable media, wherein each medium may be configured to include or includes thereon data or computer executable instructions for manipulating data. The computer executable instructions include data structures, objects, programs, routines, or other program modules that may be accessed by a processing system, such as one associated with a general-purpose computer capable of performing various different functions or one associated with a special-purpose computer capable of performing a limited number of functions. Computer executable instructions cause the processing system to perform a particular function or group of functions and are examples of program code means for implementing steps for methods disclosed herein. Furthermore, a particular sequence of the executable instructions provides an example of corresponding acts that may be used to implement such steps. Examples of computer readable media include random-access memory (“RAM”), read-only memory (“ROM”), programmable read-only memory (“PROM”), erasable programmable read-only memory (“EPROM”), electrically erasable programmable read-only memory (“EEPROM”), compact disk read-only memory (“CD-ROM”), or any other device or component that is capable of providing data or executable instructions that may be accessed by a processing system.

(47) For example, the computer device may be a personal computer, a notebook computer, a personal digital assistant (“PDA”) or other hand-held device, a workstation, a minicomputer, a mainframe, a supercomputer, a multi-processor system, a network computer, a controller, a processor-based consumer electronic device, or the like.

(48) As a result of the present invention, a mechanism for remote adjustment of the output volume

per revolution of a fixed displacement pump is provided. By extending the linear actuator, the angle and output volume per revolution of the pump is increased. By retracting the linear actuator, the angle and output volume per revolution of the pump is decreased.

(49) Moreover, by using a flexible member instead of a rigid member to link between the linear actuator and the upper base portion, a proportional relationship is established between the linear motion of the linear actuator and the angular motion of the upper base portion relative to the lower base portion. Also, the ability for electronic adjustment of flow allows the fixed displacement pump to utilize a large output volume per revolution to prime the lines, and then switch to a low output volume per revolution for the required small volume dispenses without manual intervention.

(50) The present invention further overcome issues of varying aspirate volumes relative to dispense volumes in a fixed displacement pump. Traditionally, these pumps have only been used to move fluid by rotation of the main motor. With the ability to electronically adjust the angle of the base, a new method to move fluid with a syringing motion becomes possible. With the piston flat open to one port, extending the linear actuator increases the angle of the base and pulls fluid into the pump head. In contrast, by retracting the linear actuator, the angle of the base decreases and pushes fluid out of the pump head. Furthermore, due to the flexible member, the linear motion has a proportional relation to the angular motion, which in turn has a proportional relation to the output volume. This extension and retraction gives a predictable aspirate and dispense volume from the active port.

(51) In addition, by introducing the electronic adjustment of the angle for the syringing function in valve-less pumps, it is possible to adjust the barrel size by varying the angle of the piston flat relative to the active port.

(52) Also, with the ability to both operate as a traditional syringe pump and a traditional fixed displacement pump, the variable displacement pump can prime the lines by operating like a traditional fixed displacement pump and give constant flow rate dispenses by operating like a traditional syringe pump.

(53) While various embodiments of the present invention are specifically illustrated and/or described herein, it will be appreciated that modifications and variations of the present invention may be effected by those skilled in the art without departing from the spirit and intended scope of the invention.

## Claims

1. An angle adjustment mechanism for a pump and a motor comprising: a base including an upper base portion having a first end for mounting the motor and a second end, a lower base portion for mounting the pump on a first end and a second end, and a hinge pivotably connecting said second end of said upper base portion and said second end of said lower base portion; a worm screw; an actuating motor affixed to the upper base portion and having a shaft, wherein a distal end of the shaft is connected to the worm screw; an arcuate member attached to the lower base portion and having a plurality of teeth that engage the worm screw; and an attachment plate that attaches the actuating motor to the first end of the upper base portion, wherein said attachment plate extends outwardly from the actuating motor, and wherein the actuating motor is mounted on the attachment plate, wherein actuation of said actuating motor drives said arcuate member in a curved path causing said lower base portion and upper base portion to pivot with respect to each other about said hinge, thereby changing an angle between said lower base portion and said upper base portion, and wherein the actuating motor is mounted to a top surface of the attachment plate with the shaft extending through the attachment plate, wherein a bracket is attached to and extends outwardly from a collar attached to the lower base portion, the bracket is positioned on a side of the lower base portion opposite the hinge, and wherein the bracket has two parallel members and each parallel member has a distal end, and wherein the arcuate member is disposed between and secured

to each distal end of the parallel members.

2. The angle adjustment mechanism as defined in claim 1, wherein the plurality of teeth is on a convex surface of the arcuate member facing away from the pump.

3. The angle adjustment mechanism as defined in claim 1, wherein said upper base portion comprises a flange and said flange attaches the upper base portion to the attachment plate.

4. The angle adjustment mechanism as defined in claim 1, wherein the actuator motor is a stepper motor.

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